The DICTO Ogdoecontahedron

A convex icosahedral solid of 20 regular hexagons and 60 kites: how it appeared when the Spheres of Dharma were turned inside out, the solid itself, its place in the reversed cycle, its face arrangement, and its packings



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Version 1 · 16 October 2026 · DICTO DOI: 10.5281/zenodo.23404688 (all versions) · Licence CC BY 4.0 · Data, scripts and exact checks are included with this record and at https://github.com/DICTOR-Master/kaleidohedra (branch research: studies/2026-10-16-polar-dharma, studies/2026-10-16-ocd-pyrochlore, studies/2026-10-16-ogdoecontahedron-bridge).
The Hecatodihedron, the idea of turning the Spheres of Dharma inside out (first by flipping its pieces, then by polar reciprocation), the name Ogdoecontahedron, and the idea of joining copies on their hexagons into pyrochlore networks are DICTO's. Computation, exact arithmetic and renders were done with Claude (Anthropic) as a research assistant.

Abstract. DICTO asked what the Spheres of Dharma, an exact endless nesting of six icosahedral solids, look like turned inside out. We include the cycle's in-between stage, the DICTO Truncated-Icosahedral Hecatodihedron, as a seventh station and take every station's polar reciprocal: the nesting reverses, and every station becomes a classical solid except one. The polar of the Hecatodihedron is new, and DICTO named it the **Ogdoecontahedron** for its 80 faces. It has 20 regular hexagons and 60 kites, 102 corners and 180 edges, only two edge lengths and two dihedral angles, no insphere and no midsphere, and 90 of its corners on one sphere, the inside-out image of the Hecatodihedron's 90 faces touching one sphere; every coordinate is exact in $\mathbb{Q}(\sqrt{5})$. It is the pentakis dodecahedron with its 20 six-fold corners cut off flat. In the reversed cycle its 90 sphere corners lie on the icosidodecahedron's sphere, its hexagons hold the icosidodecahedron's triangles and its kites lie in the pentakis dodecahedron's faces. Its face arrangement is known (the dual of L0 applied to the dodecahedron, in the notation of the Antiprism software), but the metric solid was not found in the sources checked. Copies joined face to face on hexagons form a diamond network whose joints lie on a pyrochlore lattice; two such networks together are a body-centred cubic packing filling exactly $(9225 - 3915\sqrt{5})/608 \approx 77.43\%$ of space. It is neither a patch nor a cluster of the canonical (primitive) icosahedral quasicrystal; five of its periodic packings meeting round a five-fold axis are the subject of a separate paper (#30).

1. How it was found

The **Spheres of Dharma** (DICTO DOI: 10.5281/zenodo.23310833) is an exact, endlessly repeating nesting of six icosahedral solids, the Ritsudō cycle: icosahedron ⊂ icosidodecahedron ⊂ dodecahedron ⊂ truncated icosahedron ⊂ rhombic triacontahedron ⊂ icosahedron φ^3 larger, each layer filled by a few repeated pieces. In its truncated icosahedron → triacontahedron layer, the hexagonal caps close part-way into a convex solid, the **DICTO Truncated-Icosahedral Hecatodihedron** (DICTO DOI: 10.5281/zenodo.23360858): the truncated icosahedron with a Shield, a low hexagonal pyramid, on each hexagon, with 102 faces (12 pentagons, 30 rhombi, 60 triangles) and 80 corners. It is the seventh stage of the cycle when each closed stage is taken as one solid (Capturing the In-betweens, DICTO DOI: 10.5281/zenodo.23345502).

On 13 October 2026 DICTO asked what the cycle looks like turned inside out. Turning each layer's pieces over does not close any layer into a shell (companion paper). **Polar reciprocation** does the job: in a sphere about the common centre it sends each face plane to a point and each corner to a plane, so every solid goes to its dual and the nesting reverses (a classical fact of convex geometry; see e.g. Firsching 2015). We took the six solids of the cycle together with its Hecatodihedron stage, seven stations in all, and reciprocated each exactly in $\mathbb{Q}(\sqrt{5})$. The reversed chain is

dodecahedron ⊂ icosidodecahedron ⊂ Ogdoecontahedron ⊂ pentakis dodecahedron ⊂ icosahedron ⊂ rhombic triacontahedron ⊂ dodecahedron φ^3 larger.

Every station is classical except the polar of the Hecatodihedron; reciprocating the six solids of the cycle alone gives classical solids only, so the new solid comes from the in-between stage. We drew it on its own, and DICTO named it for its faces, as he named the Hecatodihedron for its 102: the **Ogdoecontahedron**, from Greek ὀγδοήκοντα (eighty) and ἕδρα (face). (The #30 paper calls it the second station, counting the inner dodecahedron as station 0.) The reversed cycle as a whole, its pieces and its gaps, is the subject of a companion paper, *The Spheres of Dharma Turned Inside Out*.

2. The solid

Coordinates are in the scale of the reversed cycle (reciprocation in the unit sphere); all are exact in $\mathbb{Q}(\sqrt{5})$ unless marked float.

- **80 faces, 102 corners, 180 edges** ($102 - 180 + 80 = 2$). Each face is the dual of a Hecatodihedron corner. The 20 Shield apexes, where six faces meet, become **20 regular hexagons**; the 60 corners of the truncated icosahedron, where four faces meet, become **60 kites**. Corners: 12 where five faces meet, 30 where four meet, 60 where three meet.
- **Two edge lengths.** The hexagon side squared is $2/9 - 2\sqrt{5}/27$ (side 0.237881); each kite has two sides of that length, next to the hexagons, and two of squared length $86/3249 + 2\sqrt{5}/361$ (side 0.197124): 120 edges of the first length and 60 of the second. The hexagons are regular (all angles 120°); the kite angles are 117.85° (twice), 68.62° and 55.69°.
- **One sphere, inside out.** 90 corners lie exactly on the sphere of radius 2/3; the other 12 sit just outside, at squared radius $(170 + 2\sqrt{5})/361$. The Hecatodihedron has 90 faces touching one sphere and its 12 pentagons a little inside it: reciprocation trades a tangent sphere of faces for a sphere of corners.
- **Built from the dual side.** The Hecatodihedron adds a pyramid-like Shield to each hexagon of the truncated icosahedron; dually, the Ogdoecontahedron is the pentakis dodecahedron (the truncated icosahedron's dual) with its 20 six-fold corners **cut off flat** into regular hexagons. Adding a cap on one side is cutting a corner on the other.
- **Two face distances, no insphere.** The hexagons lie at squared distance $2/9 + 2\sqrt{5}/27$ from the centre (0.6228), the kites further out at $(42 + 2\sqrt{5})/109$ (0.6529).
- **No midsphere.** The 120 hexagon edges touch the sphere of squared radius $7/18 + \sqrt{5}/54$ (0.65597); the 60 short kite edges come no nearer than their inner ends, on the sphere of radius 2/3.
- **Two dihedral angles** (float): 156.20° on the 120 kite–hexagon edges and 156.72° on the 60 kite–kite edges, as the Hecatodihedron has only two edge angles.

- **Volume** (−1000 + 1240√5)/1539. It is convex and centrally symmetric (checked exactly), with full icosahedral symmetry.

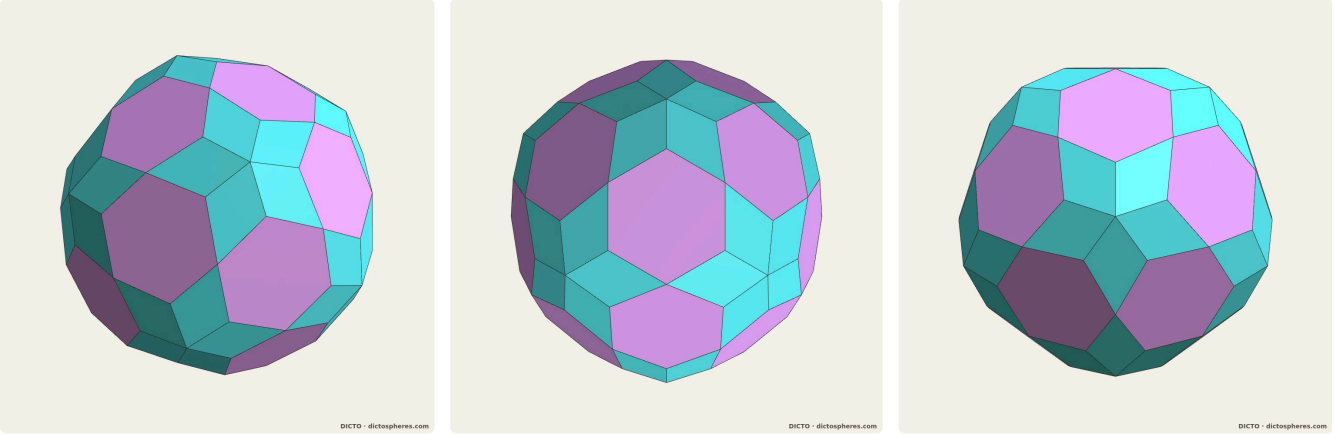


Figure 1. The DICTO Ogdoecontahedron, hexagons wisteria, kites cyan: (a) oblique; (b) along a three-fold axis, a regular hexagon in the centre; (c) along a five-fold axis, one of the 12 outer corners where five kites meet.

3. Its place in the reversed cycle

- **On the icosidodecahedron's sphere.** Its 90 sphere corners lie exactly on the icosidodecahedron's circumsphere (radius 2/3); 30 of them, its two-fold corners, are the icosidodecahedron's own corners.
- **Held between two classical solids.** Its 20 hexagons lie in the planes of the icosidodecahedron's 20 triangles, each hexagon holding a triangle; its 60 kites lie in the faces of the pentakis dodecahedron around it.

Each layer beside it is cut into one kind of piece; those pieces belong to the reversed cycle and are given in the companion paper.

4. Its face arrangement is known; the solid is not

Combinatorially (which faces meet which, ignoring shape), the Ogdoecontahedron is the pentakis dodecahedron with its 20 six-fold corners cut down to the edge midpoints. In the operator notation of the Antiprism software (Rossiter) it is the dual of L0 applied to the dodecahedron, so the Hecatodihedron has the arrangement of L0(D). This was first reported in the #30 paper (§2). We checked it by an exact isomorphism search on the face maps (mirror images allowed), with controls: randomly relabelled corners still match, and swapping two corners of one hexagon breaks the match. L0 was built from Antiprism's written definition; the program itself was not run. This agrees with the Hecatodihedron paper's own description of its arrangement, the hexakis truncated icosahedron k6tI flattened so that coplanar pairs of its triangles form the 30 rhombi.

What we did not find is this metric solid, with regular hexagons and these kites. McCooey's Visual Polyhedra (553 solids, scanned in full) has no solid with its counts or its dual's. The nearest catalogued relative is the 6-truncated pentakis dodecahedron (20 hexagons, 60 pentagons), whose truncation, carried to the edge midpoints, gives this arrangement. The printed catalogues (Wenninger, Holden, Pugh) were ruled out by scope, not read.

5. Packings: pyrochlore networks

DICTO asked whether copies joined on their hexagons form pyrochlore networks. The construction below, the five inscribed cubes and the body-centred cubic packing were first stated in the #30 paper (§3); here we give the single packings in full. Because the solid is centrally symmetric, a copy shifted by $t = 2dn/|n|^2$ (n the hexagon's normal, d its distance) puts its opposite hexagon exactly on the shared one, with no rotation. Joining through a tetrahedral set of four hexagons (pairwise cosine $-1/3$) gives a **diamond network**, with $|t|^2 = 8/9 + 8\sqrt{5}/27$. Over 60 sites (1,770 pairs) no two copies overlap (exact test: two copies overlap if and only if half their offset lies inside the solid, since $K - K = 2K$); the only contacts are full-hexagon joints. Around every copy the four joint hexagons have their centres at the corners of a regular tetrahedron, and the joints, at the diamond bond midpoints, lie on the **pyrochlore lattice** of corner-sharing tetrahedra.

A second network, shifted into the diamond's empty sites, fits with no overlap and touches the first only on hexagons. Together they are one **body-centred cubic lattice packing**: every copy in the same orientation, touching eight neighbours through the eight hexagons that point to the corners of a cube. It fills exactly $(9225 - 3915\sqrt{5})/608 \approx 77.43\%$ of space, more than the densest packing of equal spheres (74.05%). A local search over lattices (Betke–Henk style, 200 starts, float) found a denser lattice packing, 0.789260, each copy touching 12 neighbours (10 on hexagons, 2 on kites); the best of 200 local-search starts, all converging to it, checked as a packing for lattice coefficients up to 4 (float); it is not proved to be the densest.

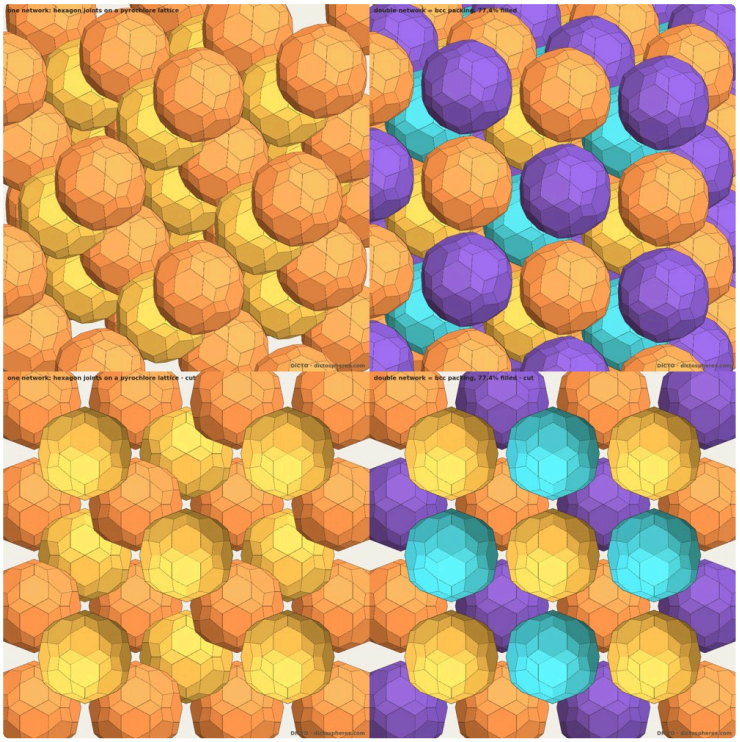


Figure 2. Hexagon-joined packings: (a) one network, its joints on a pyrochlore lattice; (b) two networks together, the body-centred cubic packing, 77.4% filled; (c), (d) the same, cut. Colours only tell copies apart.

6. Periodic and aperiodic

The Ogdoecontahedron combines the regular hexagon of periodic honeycombs with icosahedral symmetry, the symmetry of quasicrystals. We tested whether it is a piece of the canonical icosahedral quasicrystal, the primitive six-dimensional lattice cut with the canonical

(triacontahedral) window. A known case passes first: the triacontahedron spanned by the six unit vectors fits the window, with all its edges along five-fold axes.

- **Lift.** Its corners become integer six-dimensional points only after scaling by 171; their perpendicular images reach radius 116, against a window of 1.9. They first fit after nine golden inflations, when the solid's radius is about 4,750 tiling edges. Any finite golden point set fits eventually, so this says only that its corners are scattered points of a very large patch.
- **Edges.** None of its 180 edges runs along a symmetry axis, whereas every edge of the tiling runs along a five-fold axis, so it is not a cluster of the tiling.
- **Shells.** Its 30 two-fold corners are exactly the icosidodecahedron's, the 30-atom shell of the Tsai and Mackay clusters (Takakura et al. 2007): a classical tie. Its other 60 sphere corners lie 16.47° from the nearest five-fold axis, matching neither the Bergman 60-shell (20.08°) nor the Tsai 60-shell (17.78°).
- **Faces.** No corner angle of its faces (120°; 117.85°, 68.62°, 55.69°) can be made from golden-rhombus angles (63.43°, 116.57°) or Penrose angles (multiples of 36°) with up to six tiles at the corner.

So it is neither a patch nor a cluster of the canonical icosahedral quasicrystal. Other six-dimensional lattices (face- or body-centred), other windows and decorated tilings were not tested. Its 20 hexagon directions contain exactly five inscribed cubes, so it has five body-centred cubic packings. When these five periodic packings meet round a five-fold axis, the copies that must be removed along that axis, together with the central copy, form exactly the Fibonacci chain near the centre; that is the subject of a separate paper (#30, DICTO DOI: 10.5281/zenodo.23396361).

7. Relation to known work, and reproducibility

Known and credited: polar reciprocation and duality (convex geometry; Firsching 2015); the pentakis dodecahedron and the truncation of its six-fold corners; the face arrangement, as above (Rossiter; McCooey); the diamond and pyrochlore lattices and body-centred cubic packing; lattice packings of convex polyhedra and the local-search method (Betke & Henk 2000); the packing density of equal spheres (Hales 1998). **Not found** in our reference library, McCooey's catalogue and web searches (16 October 2026; the printed catalogues ruled out by scope, not read): the metric Ogdoecontahedron, with regular hexagons and kites of 117.85°, 68.62° and 55.69°; its exact hexagon-joined packings.

All scripts and data below are included with this record; in the repository they are on branch research. In studies/2026-10-16-polar-dharma: `polar.py` (exact poles of the Hecatodihedron), `hstar_render.py` and `render_hstar.html` (Figure 1). Flip test: studies/2026-10-16-flip-test, `flip.py`. In studies/2026-10-16-ocd-pyrochlore: face arrangement `ocd_dl0d.py` and `ocd_catalogue_scan.py`; networks `network.py`; densest lattice packing `densest.py`. Quasicrystal check: `bridge.py` and `bridge.txt`, studies/2026-10-16-ogdoecontahedron-bridge (also at commit 3fd407f: https://github.com/DICTOR-Master/kaleidohedra/tree/3fd407f/studies/2026-10-16-ogdoecontahedron-bridge).

References

1. M. Firsching, Computing maximal copies of polyhedra contained in a polyhedron, *Experimental Mathematics* 24 (2015) 98–105, doi:10.1080/10586458.2014.956374.

2. U. Betke, M. Henk, Densest lattice packings of 3-polytopes, *Computational Geometry* 16 (2000) 157–186, doi:10.1016/S0925-7721(00)00007-9.
3. T. C. Hales, An overview of the Kepler conjecture, arXiv:math/9811071 (1998).
4. A. Rossiter, Antiprism, polyhedron modelling software, https://www.antiprism.com.
5. D. McCooey, Visual Polyhedra, https://dmccooey.com/polyhedra (scanned 15 October 2026).
6. H. Takakura, C. P. Gómez, A. Yamamoto, M. de Boissieu, A. P. Tsai, Atomic structure of the binary icosahedral Yb–Cd quasicrystal, *Nature Materials* 6 (2007) 58–63, doi:10.1038/nmat1799.
7. DICTO, Spheres of Dharma, DICTO DOI: 10.5281/zenodo.23310833.
8. DICTO, Capturing the In-betweens, DICTO DOI: 10.5281/zenodo.23345502.
9. DICTO, The DICTO Truncated-Icosahedral Hecatodihedron, DICTO DOI: 10.5281/zenodo.23360858.
10. DICTO, The Spheres of Dharma Turned Inside Out (companion paper, in preparation).
11. DICTO, A Fibonacci Spine in a Five-fold Chorus of Ogdoecontahedron Packings, DICTO DOI: 10.5281/zenodo.23396361.